Repair the reasoning, not just the answer

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Coordinate sign / ES: signo de la coordenada / PT: sinal da coordenada

A minus sign is not "slow." Compare sizes for speed.

Also on your note page: Distance vs. displacement, Net force, Interaction pair.

B READ THE RELATIONSHIP

A sign shows direction on the axis you chose. It can change the meaning of an answer.

Find the first wrong step, repair it, then recheck

C TRACE THE ORIGINAL FIGURE

1 **CLAIM** ✗ "3 m right then 3 m back – the distance is zero."	1 **CLAIM** ✗ "8 N pull on a 2 kg cart: 8 ÷ 2 = 4 m/s²."
2 **FIX** ✓ Distance = 3 + 3 = 6 m. *Displacement* is 0 m: the endpoints match.	2 **FIX** ✓ Net force first: 8 – 2 = 6 N. Then 6 ÷ 2 = 3 m/s².
1 **CLAIM** ✗ "–2 → –6 m/s is slowing, because it's negative."	1 **CLAIM** ✗ "The force pair cancels, so nothing moves."
2 **FIX** ✓ Speed grows from 2 to 6 m/s: speeding up. The minus sign is direction.	2 **FIX** ✓ Each pair force acts on a different cart. Draw one on each diagram.
☐ **1 · PENCIL** Mistaken claims: cross out	☐ **2 · CODE COLOR** The fixes: color by quantity

Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg cart has 9 N right and 3 N left. A student writes 9 ÷ 3 = 3 m/s². Correct it.

Given m = 3 kg 9 N right 3 N left

Rule a = F net ÷ m

Work F net = 9 – 3 = 6 N right a = 6 ÷ 3

Answer 2 m/s² right missed: net force first

E SAVE FOR THE FINAL MEETING

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

Read the physics: Coordinate sign

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Write the positive direction first. Find the first step where the model, sign or starting value went wrong. Changing only the final number can hide the mistake.

02 HOW IT WORKS

When forces oppose, give them opposite signs before adding. When reading velocity, compare its size to tell whether speed grows. A more negative velocity can mean a larger speed.

03 WHAT TO CHECK

Repair the first wrong step. Then use a sketch, force sum or unit check to test the new reasoning.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg cart has 9 N right and 3 N left. A student writes $9 \div 3 = 3$ m/s². Correct it. 2 m/s² right missed: net force first.

05 CONNECT TO XJ

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

SAY IT BEFORE YOU WRITE

What failed first: the rule, the sign or the arithmetic? The first error was ___. I replaced it with ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Distance vs. displacement

ES: distancia y desplazamiento / PT: distância e deslocamento

Meaning in my words	How can you tell?
How would you explain the difference between distance and displacement? In my words, this means ___.	Which quantity can be zero after a walk that covers six metres? I can tell because ___.
Example or sketch	**Non-example and why**
Walk 4 m east and 1 m west. What distance and displacement would you record? My example shows ___.	Someone reports 5 m east as the displacement. Which part of the return trip did they miss? I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Coordinate sign

ES: signo de la coordenada / PT: sinal da coordenada

Meaning in my words	How can you tell?
How would you explain coordinate sign to a friend? In my words, this means ___.	If right is positive, what sign goes on a 3 N force left? I can tell because ___.
Example or sketch	**Non-example and why**
How would you label a leftward velocity of 2 m/s when right is positive? My example shows ___.	Does a negative velocity by itself mean the cart is slowing down? I would change ___ because ___.

Interaction pair

ES: par de interacción / PT: par de interação

Meaning in my words	How can you tell?
How would you explain interaction pair to a friend? In my words, this means ___.	Earth pulls an apple down. What is the partner force, and which object feels it? I can tell because ___.
Example or sketch	**Non-example and why**
How would you label the pair when a hand pushes a cart? My example shows ___.	Are the table's push and Earth's pull on one book a third-law pair? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

Labels you can use: first error · repair · check.

First wrong step

Repair

Tell a partner

The first error was ___. I replaced it with ___ because ___.

Stop the faulty graphic

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SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use all forces to calculate acceleration.

CHOOSE ONE JOB TOGETHER

SOFTWARE TESTER ☐

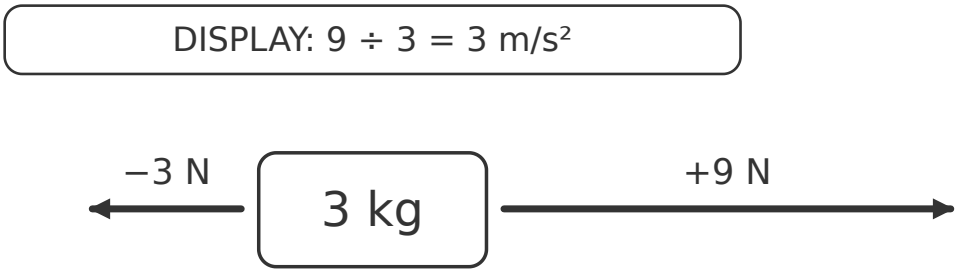
Mark the first wrong calculator step. Send the developer the corrected force sum and expected acceleration.

FACT-CHECK PRODUCER ☐

Stop or clear the graphic. Give the anchor a corrected number that viewers can check against the force arrows.

Word help: force sum / mass / acceleration

A calculator overlay for a **3 kg** tracker shows **+9 N** and **–3 N**. Its output is "**9 ÷ 3 = 3 m/s².**" It is about to go live.



01 OUR FIRST CALL

Can this calculation go live? Give the corrected result and explain.

The force sum is ___, so acceleration is ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Did the developer fix the error? Send a follow-up that names what still needs changing.

Now we would say ___ because ___.

Switch speakers. Did the calculation include both forces?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks